2.4 $\pi^2/6$

WORKSHEET

The question

(i) Use De Moivre's theorem to show that, if $\sin\theta \neq 0$, then

$$\frac{(\cot\theta + \mathrm{i})^{2n+1} - (\cot\theta - \mathrm{i})^{2n+1}}{2\mathrm{i}} = \frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta},$$

for any positive integer n.

Deduce that the solutions of the equation

$$\binom{2n+1}{1}x^n - \binom{2n+1}{3}x^{n-1} + \cdots + (-1)^n = 0$$

are

$$x = \cot^2\left(\frac{m\pi}{2n+1}\right)$$

where $m = 1, 2, \ldots, n$.

(ii) Hence show that

$$\sum_{m=1}^{n} \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{n(2n-1)}{3}.$$

(iii) Given that $0 < \sin\theta < \theta < \tan\theta$ for $0 < \theta < \frac{1}{2}\pi$, show that

$$\cot^2\theta < \frac{1}{\theta^2} < 1 + \cot^2\theta.$$

Hence show that

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6}.$$

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If a polynomial $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has roots $r_1, \ldots, r_n$, what is $r_1 + \cdots + r_n$ in terms of the coefficients?

STAGE 1 De Moivre

Ask yourself Write $\cot\theta + \mathrm{i}$ as a single fraction. What do you notice about the numerator?

STAGE 2 The polynomial

Ask yourself In $(c+\mathrm{i})^N - (c-\mathrm{i})^N$, which terms of the two binomial expansions cancel, and which double?

Pause and try For which θ is the left-hand side zero? Use this to find n different roots of the polynomial.

STAGE 4 The squeeze

Ask yourself Put $\theta = \frac{m\pi}{2n+1}$ and add up over $m = 1, \ldots, n$. What do the two outer sums become? What does the middle sum have to do with $\sum 1/m^2$?

Try this next

1 **The 1992 version.** (From STEP III, 1992.) Using the polynomial from part (i), show that

$$\sum_{k=1}^{n} \tan^2\left(\frac{k\pi}{2n+1}\right) = n(2n+1) \quad \text{and} \quad \sum_{k=1}^{n} \operatorname{cosec}^2\left(\frac{k\pi}{2n+1}\right) = \frac{2n(n+1)}{3}.$$

(*Hint for the first:* the numbers $\tan^2$ are the reciprocals of the roots. What polynomial has roots $1/r_i$?)

2 **Odd squares.** Deduce from $\sum 1/m^2 = \pi^2/6$ that $1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \cdots = \frac{\pi^2}{8}$.

3 **Fourth powers.** HARDER Use $\sum r_i^2 = (\sum r_i)^2 - 2\sum_{i<j} r_i r_j$ and the coefficients of the polynomial to find $\sum_{m=1}^{n} \cot^4\left(\frac{m\pi}{2n+1}\right)$, and adapt the squeeze to prove $\sum 1/m^4 = \pi^4/90$.

4 **How fast?** Using the lower bound in part (iii), show that $\frac{\pi^2}{6} - \sum_{m=1}^{n} \frac{1}{m^2} < \frac{\pi^2}{4n}$. How large an n does this guarantee is enough for the partial sum to be within 0.001 of $\pi^2/6$? Compare with the true error, which is close to $1/n$.

Further reading

- **William Dunham, *Euler: The Master of Us All*** (Mathematical Association of America). Chapter 3 tells the story of the Basel problem with the mathematics in full.
- **Aigner & Ziegler, *Proofs from THE BOOK***, Chapter 9, "Three times $\pi^2/6$". Includes the proof in this section.
- **Paul Nahin, *Dr. Euler's Fabulous Formula*** (Princeton University Press). Complex numbers, Fourier series and sums like this one.

The full guided solution, and what the question leads to at university, is in Section 2.4:
https://keithmcnulty.github.io/step_book_mcnulty/ch02/basel.html